

MedTrack – Milestone 2

Member 6: Dashboard Storyboard

Storyboard structure: For each dashboard — Purpose → KPIs → Charts → Filters → Interactions

Dashboard 1: Hospital Overview

Purpose: Provide hospital management with a quick summary of hospital activity, patient admissions, bed availability, and overall operational performance.

KPIs

Total Admissions • Occupancy Rate • Average Length of Stay • Readmission Rate • Bed Utilization Rate

Charts

Chart	Type	What it shows
Admissions Trend	Line chart	Admission Date vs. Number of Admissions
Admissions by Department	Horizontal bar chart	Department vs. Number of Admissions
Bed Status	Donut chart	Occupied vs. Available Beds
Admissions by Admission Type	Bar chart	Elective vs. Emergency Admissions
Average LOS by Department	Bar chart	Department vs. Average Length of Stay

Filters

Year • Admission Date • Department • Ward • Admission Type • Disease • Admission Status

Interactions

- Selecting a department filters the related charts and KPI information.
- Changing the year updates the KPIs and time-based charts.
- Selecting an admission type updates admission-related visuals.
- Selecting a ward updates relevant bed and admission information.
- Clicking a department cross-filters the dashboard.

Dashboard 2: Patient & Clinical Performance

Purpose: Analyze patient admissions, diseases, length of stay, and readmission patterns to understand clinical activity and patient outcomes.

KPIs

Total Admissions • Average Length of Stay • Readmission Rate

Charts

Chart	Type	What it shows
Admissions by Department	Horizontal bar chart	Department vs. Number of Admissions
Average LOS by Department	Bar chart	Department vs. Average Length of Stay
Readmission Rate by Department	Bar chart	Department vs. Readmission Rate
Admissions by Disease Category	Bar chart	Disease Category vs. Number of Admissions
Admissions by Admission Type	Donut chart	Elective vs. Emergency Admissions

Filters

Year • Admission Date • Department • Ward • Disease • Disease Category • Admission Type

Interactions

- Selecting a disease category filters disease and admission visuals.
- Selecting a department updates the clinical performance charts.
- Selecting an admission type updates related KPIs and charts.
- Changing the year enables comparison of patient activity across periods.
- Clicking a disease category cross-filters the dashboard.

Dashboard 3: Bed & Department Performance

Purpose: Monitor hospital bed capacity and compare department operational performance and resource utilization.

KPIs

Total Beds • Occupied Beds • Available Beds • Occupancy Rate • Bed Utilization Rate • Department Efficiency Score

Charts

Chart	Type	What it shows
Bed Status	Stacked bar chart	Occupied vs. Available Beds
Beds by Ward	Bar chart	Ward vs. Total Beds
Bed Utilization by Department	Horizontal bar chart	Department vs. Bed Utilization Rate
Department Efficiency Score	Ranked horizontal bar chart	Department vs. Efficiency Score
Admissions vs. Bed Capacity	Scatter plot	Total Beds vs. Total Admissions by Department

Filters

Year • Department • Ward • Ward Type • Admission Type • Admission Date

Interactions

- Selecting a department filters the ward and bed information.
- Selecting a ward updates bed-status information.
- Clicking a department in the efficiency chart filters the other charts.
- Changing the year updates admissions and utilization-related information.
- Hovering over charts displays department, ward, or KPI details.

Dashboard Navigation & Overall Story

Navigation: Dashboard 1 → Dashboard 2 → Dashboard 3, using dashboard buttons or a top navigation menu.

Dashboard	Main Question
Hospital Overview	What is happening in the hospital?
Patient & Clinical Performance	What is happening with the patients and clinical activity?
Bed & Department Performance	How efficiently are hospital resources and departments being used?

Dataset Fields Used

- **Admission:** admission_date, discharge_date, admission_type, admission_status, patient_id, department_id, ward_id, bed_id, disease_id
- **Bed:** bed_id, bed_number, bed_status, ward_id
- **Department:** department_id, department_name, department_type, floor_number, status
- **Ward:** ward_id, ward_name, ward_type, total_beds, department_id
- **Disease:** disease_id, disease_name, disease_category
- **Patient:** patient_id, gender, date_of_birth, blood_group, city
- **Employee:** employee_id, role, employment_type, department_id

Note: This document is the Member 6 dashboard storyboard deliverable. It describes the dashboard design and user interactions; it does not cover the implementation work assigned to other members.